

Bryn Mawr College
Econ 311: Game Theory and Applications
Fall Semester 2025

Contact Information:

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This syllabus provides a general plan for the course; deviations may be necessary.

Course Objective:

Game theory studies interactions between people, corporations, institutions, or countries in which each player recognizes their strategic interdependence with the others in the game. Many economic decisions in the real world have such strategic interdependence. The course uses theoretical models and techniques from game theory to examine economic incentives in various strategic scenarios. Applications may include oligopoly, zero and nonzero-sum games, cooperative and non-cooperative games, asymmetric information, adverse selection, signaling, and bargaining models.

Prereq. ECON 200 - Intermediate Microeconomics.

Textbooks

- *Game Theory and Behavior* by Jeffrey Carpenter and Andrea Robbett (MIT Press).
- *Strategy and Game Theory: Practice Exercises with Answers* by Felix Munoz-Garcia Daniel Toro-Gonzalez (There is an electronic copy available in the library) - [Link](#)
- *Strategy: An Introduction to Game Theory* by Joel Watson
- *Introduction to Game Theory* by Martin J. Osborne

Course Requirements:

The course requirements include homework assignments and three exams. Homework will be submitted on Moodle or in-class and must be clearly presented for full credit. Exams 1 and 2 will be in in class.

Assessment	Date	Share of final grade
Class participation		10%
Exam 1	October 6, 2025	25%
Exam 2	December 3, 2025	25%
Homework		15%
Proposal presentation	October 8, 2025	5%
Project		15%
Final presentation	December 8 & 10, 2025	5%

A numeric grade of 4.0, 3.7, 3.3, 3.0, 2.7, 2.3, and 2.0 will be assigned to you on the basis of your cumulative score.

4.0: 95%-100% 3.7: 90%-94% 3.3: 87%-89% 3.0: 83%-86% 2.7: 80%-82% 2.3: 77%-79% 2.0: below 76%.

Homework

Homework will be due no less than one week after they are assigned. You must turn in your own work, but group effort is permitted and encouraged. Homework turned in late after the due date will be docked 10 percentage points every 24 hours. After 5 days late, the maximum possible grade will be 50% of the full points.

Research Project

The research project provides an opportunity to apply game theory concepts and analytical tools to examine real-world strategic interactions. Students will conduct independent research analyzing a strategic situation using appropriate game-theoretic models and solution concepts covered in the course.

Project Scope: Students will identify a strategic scenario involving multiple decision-makers where game theory provides valuable insights. This could include economic markets, political processes, social phenomena, organizational behavior, or policy issues. The analysis should demonstrate understanding of game-theoretic concepts such as equilibrium analysis, strategic dominance, information asymmetries, or dynamic interactions.

Independent Research Process

Each student will conduct individual research on their chosen topic. As part of this process, consider developing a research question that examines strategic behavior in a real-world context. It's advisable to document responses to the provided questions, continually refining them through reflection, discussions with colleagues, and consultation with the instructor. Feel free to submit

your preliminary ideas at any point in their development, and don't hesitate to reach out for discussions and guidance.

1. What specific strategic interaction do you aim to analyze? (Present your response as a clear research question concluding with a question mark.)
2. What existing knowledge do you possess regarding the strategic behavior and outcomes in this situation?
3. What game-theoretic concepts and solution methods are most appropriate for analyzing this strategic interaction?
4. Assess the significance of this analysis. Evaluate whether the game-theoretic insights are likely to be surprising, policy-relevant, or capable of changing how we understand this strategic situation.

Disability Support Services: Reasonable Accommodations

Bryn Mawr College is committed to providing equal access to students with a documented disability. Students needing academic accommodations for a disability must first register with Access Services. Students can call 610-526-7516 to make an appointment with the Director of Access Services, Deb Alder, or email her at dalder@brynmawr.edu to begin this confidential process. Once registered, students should schedule an appointment with the professor as early in the semester as possible to share the verification form and make appropriate arrangements. Please note that accommodations are not retroactive and require advance notice to implement. More information can be obtained at the Access Services website. (<http://www.brynmawr.edu/access-services/>)

Any student who has a disability-related need to record this class first must speak with the Director of Access Services and to me, the instructor. Class members need to be aware that this class may be recorded.

Academic Honesty

Students are responsible for knowing and adhering to the Bryn Mawr Honor Code. If you have any questions about expectations, please ask.

Title IX:

Bryn Mawr/Haverford College is committed to fostering a safe and inclusive living and learning environment where all can feel secure and free from harassment. All forms of sexual misconduct, including sexual assault, sexual harassment, stalking, domestic violence, and dating violence are violations of Bryn Mawr/Haverford's policies, whether they occur on or off campus. Bryn Mawr/Haverford faculty are committed to helping to create a safe learning environment for all students and for the College community as a whole. If you have experienced any form of gender or sex-based discrimination, harassment, or violence, know that help and support are available. Staff members are trained to support students in navigating campus life, accessing health and counseling services, providing academic and housing accommodations, and more.

The College strongly encourages all students to report any incidents of sexual misconduct. Please be aware that all Bryn Mawr/Haverford employees (other than those designated as confidential resources such as counselors, clergy, and healthcare providers) are required to report information about such discrimination and harassment to the [Bi-College Title IX Coordinator](#).

Information about the College's Sexual Misconduct policy, reporting options, and a list of campus and local resources can be found on the College's website:

[Bryn Mawr Sexual Misconduct Policy](#)
[Haverford Sexual Misconduct Policy](#)

Tentative course outline:

Date	Topic
Monday, September 1, 2025	Labor Day - No Class
Wednesday, September 3, 2025	What is a game? What are the assumptions? How do we represent it?
Monday, September 8, 2025	Solving games - dominance (pure strategy, strong and weak)
Wednesday, September 10, 2025	Equilibrium - maximin/minimax, best reply, dominance vs. best reply
Monday, September 15, 2025	Pure strategy Nash equilibrium (2- and 3-player games)
Wednesday, September 17, 2025	Mixed strategy Nash
Monday, September 22, 2025	Equilibrium in non-matrix games: Bertrand, Cournot, social dilemmas
Wednesday, September 24, 2025	Subgame perfection: backward induction and perfect equilibrium
Monday, September 29, 2025	Subgame perfection in non-matrix games
Wednesday, October 1, 2025	Review
Monday, October 6, 2025	Exam 1
Wednesday, October 8, 2025	Proposal Presentations
Monday, October 13, 2025	Fall Break - No Class
Wednesday, October 15, 2025	Fall Break - No Class
Monday, October 20, 2025	Repeated games: finitely and infinitely repeated games
Wednesday, October 22, 2025	Repeated games continued - Folk theorem
Monday, October 27, 2025	Simultaneous games with incomplete information - Bayesian games
Wednesday, October 29, 2025	Bayesian games continued
Monday, November 3, 2025	Sequential games with incomplete information - signaling
Wednesday, November 5, 2025	Signaling continued
Monday, November 10, 2025	Auction, Voting, and Rank-choice - Mechanism design
Wednesday, November 12, 2025	Mechanism design continued
Monday, November 17, 2025	Behavioral Game Theory, Learning, Evolutionary Game Theory
Wednesday, November 19, 2025	Behavioral Game Theory continued
Monday, November 24, 2025	Learning and Evolutionary Game Theory
Wednesday, November 26, 2025	Thanksgiving Break - No Class
Monday, December 1, 2025	Review
Wednesday, December 3, 2025	Exam 2
Monday, December 8, 2025	Presentations
Wednesday, December 10, 2025	Presentations